

Excel Dashboard Portfolio Project

1. Project Overview

This document defines the detailed requirements for developing a portfolio-ready Excel dashboard project using a public dataset from Kaggle. The goal of the project is to demonstrate strong Excel skills in data cleaning, analysis, visualization, and dashboard design, suitable for data analyst roles.

The final output should be a professional, interactive Excel dashboard that tells a clear business story and supports data-driven decision-making.

2. Project Objectives

The project should demonstrate the following competencies:

- Ability to work with real-world, messy data
- Data cleaning and preparation using Excel
- Use of formulas and functions for analysis
- Creation of pivot tables and pivot charts
- Design of an interactive, user-friendly dashboard
- Business insight generation and storytelling

3. Dataset Requirements (Kaggle)

3.1 Source

- Dataset **must be downloaded from Kaggle**
- Dataset should be publicly available and free to use
- Kaggle dataset link must be included in the final documentation

3.2 Dataset Characteristics

The dataset should:

- Contain **at least 5,000 rows** (preferred) or enough data to show trends
- Include both **categorical and numerical variables**
- Represent a realistic business or real-world scenario

3.3 Recommended Dataset Domains (Choose One)

Examples include:

- Sales / Retail transactions
- E-commerce orders
- Healthcare data (appointments, outcomes, costs, etc.)
- Finance (expenses, revenue, investments)
- Customer behavior or churn
- HR or employee performance
- Supply chain or logistics

3.4 Example Kaggle Datasets (Illustrative)

- Superstore Sales Dataset
- E-commerce Sales Dataset

- Healthcare Appointment No-Shows
- Global Retail Sales Dataset

(Exact dataset selection is flexible but must meet the above criteria.)

4. Tools & Technologies

- **Microsoft Excel** (Microsoft 365 preferred)
- Features to be used:
 - Pivot Tables
 - Pivot Charts
 - Slicers
 - Timelines (if date data exists)
 - Excel formulas and functions

No external tools (Power BI, SQL, Python) should be used for analysis.

5. Data Preparation Requirements

5.1 Raw Data Sheet

- Original dataset must be stored in a sheet named: Raw_Data
- No transformations should be applied directly to the raw data

5.2 Data Cleaning Tasks

Create a separate sheet named Data_Cleaning where the following are handled:

- Removal of duplicates (if any)
- Handling missing values (remove, replace, or flag)
- Correcting data types (dates, numbers, text)
- Standardizing categorical values (e.g., country names, product categories)
- Creating derived columns if needed (e.g., Month, Year, Profit Margin)

All cleaning steps should be clearly structured and traceable.

6. Data Analysis Requirements

6.1 Analysis Sheet

Create a sheet named Analysis that includes:

- Pivot tables answering key business questions
- Summary metrics and aggregations

6.2 Mandatory KPIs

At minimum, include:

- Total value metric (e.g., Total Sales / Total Revenue / Total Cost)
- Average metric (e.g., Average Order Value, Average Cost)
- Count metric (e.g., Number of Orders, Customers, Records)
- Time-based trend (Monthly / Quarterly / Yearly)

6.3 Example Business Questions

Depending on the dataset, answer questions such as:

- How is performance changing over time?
- Which category/region/customer segment performs best?
- What are the top and bottom contributors?
- Are there seasonal or monthly trends?

7. Dashboard Requirements

7.1 Dashboard Sheet

- Create a dedicated sheet named Dashboard
- The dashboard must fit **within one Excel screen** (no excessive scrolling)

7.2 Mandatory Dashboard Components

KPI Cards

At least 4 KPI cards displaying:

- Total metric (e.g., Sales)
- Average metric
- Count metric
- One business-specific KPI

Visualizations

Include **at least 5 charts**, such as:

- Line chart (time trend)
- Bar or column chart (category comparison)
- Pie or donut chart (distribution)
- Top 5 / Bottom 5 chart
- Regional or segment-wise breakdown

Interactivity

- Minimum **3 slicers** (e.g., Date, Category, Region)
- All visuals must respond to slicer selections

8. Design & Usability Guidelines

- Consistent color theme (maximum 3–4 colors)
- Clear titles and axis labels
- Use readable fonts and proper spacing
- Avoid clutter and unnecessary gridlines
- Align visuals neatly using Excel alignment tools

The dashboard should look **professional and presentation-ready**.

9. Documentation Requirements

Create a sheet named Project_Overview containing:

- Project title
- Dataset source (Kaggle link)
- Business problem statement
- KPIs explained in simple terms
- Key insights (3–5 bullet points)
- Assumptions or limitations

10. Final Deliverables

The final Excel file should include the following sheets:

1. Project_Overview
2. Raw_Data
3. Data_Cleaning
4. Analysis
5. Dashboard

11. Portfolio Evaluation Criteria

The project will be evaluated based on:

- Data cleaning quality
- Correct use of Excel formulas and pivots
- Dashboard clarity and aesthetics
- Business insight generation
- Interactivity and usability
- Overall professionalism

12. Expected Outcome

Upon completion, this project should be suitable for:

- Uploading to a portfolio (Google Drive / GitHub / Notion)
- Showcasing during interviews
- Demonstrating Excel proficiency to recruiters and hiring managers